YEAR 8 MATHS · CHAPTER 8 · MIND MAP

Probability and statistics

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1 Interpreting graphs and tables

Read the title, the axes, the scale and the key first.

Column graphs for categories, line graphs for change over time, pie charts for parts of a whole

2 Frequency tables

Tally each result, then count. Frequency is how many times it happened.

5 out of 20 results: relative frequency $\frac{5}{20} = 25\%$

3 Graphs of frequency tables

Histograms show numerical data: the columns touch.

Also: dot plots, and stem-and-leaf plots (key 2 | 4 = 24)



4 Measures of centre

Mean: add and divide. Median: middle value in order. Mode: most common.

1, 3, 3, 6, 7: mean $\frac{20}{5} = 4$, median 3, mode 3

2, 4, 6, 9: median $\frac{4+6}{2} = 5$

5 Measures of spread

Range = max − min. IQR $= Q_3 - Q_1$, the spread of the middle half.

2, 4, 5, 7, 8, 10: range 8

$Q_1 = 4,\ Q_3 = 8$, so IQR $= 4$

Split the data in half at the median, then find the median of each half.

6 Surveying and sampling

A sample should represent the population. Random samples avoid bias.

Census: everyone. Sample: some

Asking only your friends isn't a random sample

7 Probability

$P = \frac{\text{favourable}}{\text{total}}$, from 0 (impossible) to 1 (certain).

One die: $P(5 \text{ or } 6) = \frac{2}{6} = \frac{1}{3}$

$P(\text{not } 6) = 1 - \frac{1}{6} = \frac{5}{6}$

8 Two-step experiments

List every outcome in a table.

Two coins: HH, HT, TH, TT. $P(\text{one head}) = \frac{2}{4} = \frac{1}{2}$

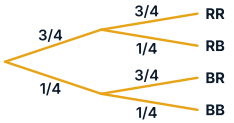
Two dice: $P(\text{sum of } 12) = \frac{1}{36}$

9 Tree diagrams

Multiply along the branches, then add the results you want.

3 red and 1 blue, with replacement:

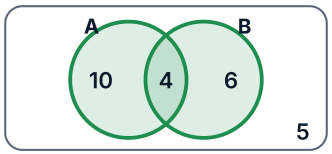
$P(\text{red, red}) = \frac{3}{4} \times \frac{3}{4} = \frac{9}{16}$



10 Venn diagrams and two-way tables

Fill in the overlap first.

25 people: $P(\text{both}) = \frac{4}{25}$



11 Experimental probability

Relative frequency $= \frac{\text{times it happened}}{\text{number of trials}}$

18 heads in 50 tosses: $\frac{18}{50} = 0.36$. The expected probability is 0.5.

More trials usually get closer to the expected probability.